AmirAbbas Ranjbar

Senior Software Engineer

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SUMMARY

A business-driven Senior Software Engineer and problem solver with 8+ years of experience designing and delivering high-throughput, scalable enterprise systems.

Played a key role in large-scale digital transformation initiatives, leading the design and implementation of mission-critical, event-driven, and streaming architectures processing 200K+ transactions per hour in cloud-native environments (AWS, Azure, Kubernetes). Experienced in building resilient distributed systems where performance, reliability, and consistency were non-negotiable. Proven ability to operate under high-traffic conditions, optimize data pipelines, and implement real-time streaming solutions capable of processing millions of records efficiently and without duplication.

I actively leverage AI-assisted development in my workflow and have incorporated agentic AI approaches to accelerate product development cycles, improve experimentation speed, and reduce time-to-delivery while maintaining engineering quality.

A committed team contributor who actively supports peers in problem-solving while remaining hands-on in development and architecture decisions.

EMPLOYMENT HISTORY:

Senior Software Engineer, Veera Smart Solution**,** *Vancouver, BC* — **May 2025 - Present**

- Designed and implemented a scalable backend architecture enabling workflow automation across accounting, resource management, facility operations, and internal service processes — reducing manual coordination and improving operational visibility.
- Built an AI-powered Trip Planner solution for the tourism sector, enabling hotels to generate personalized tour recommendations for guests while supporting managers in curating optimized tour packages.
- Leveraged AI-assisted development and agentic AI workflows to accelerate product iteration, reduce development time, and move from idea to deployable features significantly faster.
- Played a key role in bridging business and engineering by translating evolving client requirements into scalable technical architectures. Led stakeholder communications, shaped product direction, and drove the development of AI-powered features that accelerated product innovation and time-to-market.

Senior Software Engineer, Okala**,** *Tehran, Iran* — **May 2021 – May 2025**

- Designed and delivered a Real-Time Product Catalog & Pricing Engine using Kafka and Apache Flink, composed of 8 domain-driven microservices for inventory, pricing, and product availability. Replaced a 30-minute monolithic sync process with a fault-tolerant streaming architecture capable of processing up to 2 million item updates per minute in near real-time. This transformation enabled dynamic pricing for Sales & Marketing and generated profit equivalent to the company's entire annual budget — marking the first profitable cycle.
- Contributed to the modernization of a legacy monolithic system (C#/.NET) by extracting and redesigning key domains into scalable microservices built with CQRS patterns and isolated databases (SQL Server, Redis). Leveraged event-driven architecture to eliminate heavy database locking and reduce vendor service-radius update latency from ~5 minutes to near real-time. This transformation enabled integration with multiple delivery providers, introduced advanced vendor-level controls, and expanded platform capacity from 200 to over 10,000 vendors — significantly increasing service coverage and operational scalability.
- Helped drive a 70% increase in sales by improving vendor visibility, expanding customer coverage, and enhancing delivery capabilities across the platform.
- Collaborated within a cross-functional engineering team to balance large-scale architectural refactoring with ongoing production demands and business continuity.

Software Engineer, DPI Iran (ex-IBM), *Tehran, Iran* — **Apr 2019 – May 2021**

- Contributed to the modernization of a legacy urban electricity management system by launching electronic billing as a key digital transformation milestone. The shift from paper-based invoices to e-billing reduced operational costs, eliminated large-scale printing overhead, and supported environmental sustainability by significantly reducing paper consumption.
- Improved system performance by analyzing and optimizing complex SQL queries, reducing response times, and improving overall system stability. Database refactoring and modernization efforts, enhancing maintainability and preparing the platform for future upgrades.
- Containerized legacy services using Docker and established CI/CD pipelines to automate build, test, and deployment processes. Improved deployment consistency, reduced manual release overhead, and enabled faster, more reliable delivery cycles.

SKILLS

Languages & Frameworks: C#, .NET (Core & Framework), ASP.NET, WPF, Entity Framework, Node.js, Python (familiar), Go (familiar)

Backend & Architecture: Enterprise Services, Development, Service-Oriented Architecture (SOA), RESTful APIs, JSON, Design Patterns, Dependency Injection, Distributed Systems, DDD, High-Availability Systems, Scalability Engineering, Event-Driven Architecture, Apache Kafka

Concurrency & Performance: Multi-threaded Programming, Async/Await, Parallel Processing, Thread Safety, Performance Optimization, Concurrency Control

Data & ORM: SQL Server, T -SQL, Entity Framework, NHibernate, Object-Relational Mapping (ORM), Relational Database Design, Query Optimization

Testing & Quality: Test-Driven Development (TDD), Unit Testing (xUnit, NUnit), Integration Testing, Continuous Integration (CI), Code Reviews, Clean Code Principles

DevOps & Deployment: CI/CD Pipelines, Docker, kubernetes, Deployment Automation, Production Monitoring, Incident Support

Methodologies: Agile / Scrum, Cross-Functional Collaboration, User Story Refinement, Technical Documentation, SDLC

EDUCATION

Azad University (IAU) - Bachelor's degree, Computer Software Engineering | 2013 - 2018

LANGUAGES

English: Professional Working Proficiency
Persian (Farsi): Native